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WHY COSMOLOGICAL MEASUREMENTS DISAGREE

A channel-weighted account of measurement, calibration, and inference

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Abstract

Different cosmological measurements can report different values for the same named parameter because the parameter is not read directly from the universe. It is reconstructed through a complete chain: a physical signal is produced, transported, calibrated, compared with a model, combined with priors and nuisance parameters, and compressed into a familiar coordinate such as H_0 . This paper formalizes that chain using three physical channels—Light, Boundary, and Matter—and proves the first-order equation that maps a universal physical deformation into a pipeline-dependent inferred parameter. The result explains why the same instrument can produce different channel mixtures, why calibration can dominate an inference without being a raw datum, why negative cross-response is legitimate, and how the proposal can be falsified across BAO, full-shape clustering, CMB, supernova, lensing, and standard-siren pipelines.

1. The ordinary-language problem

When two collaborations quote different values of H_0 , it is tempting to imagine that each instrument has directly measured a different expansion rate. That is not what the analysis does. A telescope records photons, spectra, angles, redshifts, shapes, or timing. H_0 is produced later by an inference pipeline.

The pipeline may need a standard candle, a standard ruler, an acoustic model, a matter-transfer scale, a galaxy-bias model, a covariance matrix, external priors, and a parameterized cosmology. The final number is therefore a coordinate of the complete measurement relation, not a raw property printed on a detector.

KEY IDEA

The correct comparison is not instrument versus instrument. It is inference pipeline versus inference pipeline.

2. Three channels

Unified Mechanics compresses the relevant physical content into three channels.

Compressed object	Traditional meaning
L — Light	Anything that carries a signal outward: photon or wave propagation, redshift transport, luminosity, angular position, and direct signal transmission.
B — Boundary	The relation that converts one kind of information into another: geometry, lensing, calibration, ruler-to-distance conversion, degeneracy breaking, and cross-response.
M — Matter	Retained structure: clustering, growth, velocities, the equality scale, mass distribution, and the physical ruler written into matter.

A real observation is rarely pure. BAO uses a feature in the matter distribution, reads it through light and redshift, and interprets it through geometry. Weak lensing reads matter through its effect on light paths. Supernovae provide relative distances through light, but an absolute H_0 requires a luminosity calibration. Full-shape clustering gains power when matter-shape information calibrates the geometric distance scale.

3. Why DESI is not one channel

DESI is an instrument and survey, not a single observation operator. Its BAO analysis, full-shape analysis, redshift-space-distortion analysis, and future higher-order analyses use different combinations of the same catalogue. The DESI full-shape programme explicitly models clustering shape and redshift-space distortions, while BAO isolates distance information from the acoustic feature. Those outputs must be assigned separate response maps.

This distinction matters because saying “DESI is matter dominated” or “BAO is light dominated” is too coarse. The channel signature belongs to the exact data vector, covariance, calibration assumptions, nuisance model, priors, and estimator.

4. The data and the model

For a pipeline p , write its observed data vector as d_p . The theory predicts a mean vector μ_p . Two types of coordinates enter that prediction.

ordinary fitted parameters: $\theta = (H_0, \Omega_m, \omega_b, n_s, \dots)$

physical channel deformations: $s = (s_L, s_B, s_M)$

The first list contains the coordinates used by the conventional analysis. The second list represents changes in the underlying Light, Boundary, and Matter realization.

$$J_p = \partial \mu_p / \partial \theta$$

$$K_p = \partial \mu_p / \partial s$$

J_p asks how the model prediction moves when the fitted parameters move. K_p asks how the predicted observation moves when the physical channel realization moves. A conventional analysis knows J_p . Unified Mechanics must derive K_p from its action.

5. Why simple percentages fail

One can whiten the channel responses using the covariance matrix and expand their squared length. This produces pure-Light, cross, and pure-Matter terms. The cross-term looks exactly like Boundary.

$$\|k_L + k_M\|^2 = \|k_L\|^2 + 2\langle k_L, k_M \rangle + \|k_M\|^2$$

However, the cross-term can be negative. Increasing H_0 and increasing the matter density may move a BAO observable in opposite directions. Negative cross-response is a normal sign of cancellation or parameter degeneracy. It is not a negative amount of physical stuff.

Therefore the response terms cannot simply be normalized and called percentages. They are oriented vectors in likelihood space.

6. Boundary as information created by relation

There is another decomposition that is positive and intuitive. Ask how much information about H_0 is present in geometric data alone, how much is present in matter data alone, and how much appears only after the two are combined.

$$\mathcal{B} = \mathcal{I}_H(\text{geometry} + \text{matter}) - \mathcal{I}_H(\text{geometry}) - \mathcal{I}_H(\text{matter})$$

If neither block can determine H_0 by itself, but their combination can, the inference is Boundary dominated. The target information exists in the relation.

This is exactly what standard-ruler analyses do. The observed distance pattern does not determine an absolute expansion scale until the physical ruler is calibrated. The ruler does not determine an expansion history until it is compared with the observed geometry. H_0 emerges from their relation.

7. The solved inference equation

Assume that the physical channel state changes slightly by δs . The predicted data move by

$$\delta d_p = K_p \delta s$$

The conventional analysis does not fit δs . It changes its familiar parameters until the model best absorbs the shifted data. In the local Gaussian approximation, the parameter displacement is

$$\delta \hat{\theta}_p = F_p^{-1} J_p^T C_p^{-1} K_p \delta s$$

C_p is the data covariance. F_p is the curvature of the full likelihood, including priors and nuisance parameters. This equation is standard parameter-bias mathematics, but here it becomes the explicit compiler for Observation.

For H_0 , select the Hubble coordinate:

$$\delta \hat{H}_p = e_H^T F_p^{-1} J_p^T C_p^{-1} K_p \delta s$$

Everything specific to the pipeline is contained in the row vector

$$W_p = e_H^T F_p^{-1} J_p^T C_p^{-1} K_p$$

$$\delta \hat{H}_p = W_p \delta s$$

8. Inference shear

Now compare two pipelines p and q observing the same universe. The physical deformation δs is the same. Their projection rows differ.

$$\hat{H}_p - \hat{H}_q = (W_p - W_q) \delta s$$

This is inference shear. It says that a disagreement between reported parameters can arise from the difference between two complete inverse problems.

WHAT THE EQUATION DOES NOT ALLOW

It does not allow a new free correction for every dataset. One universal δs must pass through independently calculated W rows and predict all probe shifts.

9. Where calibration enters

A prior or calibration changes the likelihood curvature F_p . It therefore changes W_p and changes the reported parameter even when the raw data are unchanged. This is not evidence that the result is arbitrary. It means the reported coordinate belongs to the combined system of data and calibration.

Examples include the sound horizon in BAO, absolute luminosity in the distance ladder, recombination physics in CMB inference, galaxy bias in clustering, and source-redshift calibration in lensing.

In the compressed language, calibration is often Boundary: it is the relation that makes one data system commensurable with another.

10. Natural consequences

Same instrument, different O

DESI BAO, DESI full shape, and DESI redshift-space analyses are different observation operators even when they share targets and hardware.

Calibration is physical

A calibration prior is not an external bookkeeping footnote. It changes F and therefore changes the channel-to-parameter map W .

Boundary can dominate without being a raw datum

A parameter may be almost unidentifiable in either data block alone and become sharply determined only through their relation.

Negative response is meaningful

Opposing channel derivatives describe cancellation or degeneracy direction; they are not unphysical negative probabilities.

No universal H_0 outside a chart

The invariant expansion history may be unique while a conventional scalar H_0 is a coordinate returned by a specified realization and inference pipeline.

Cross-probe prediction is mandatory

Once δs is fixed, every additional W yields a prediction. Dataset-specific δs values would destroy the theory.

The best probes are complementary

A strategically chosen set of pipelines with linearly independent W rows can reconstruct the channel deformation itself.

Parameters become charts

The invariant object is the realized cosmological history. H_0 , Ω , S_8 , w_0 , and related scalars are coordinates returned by particular observation-and-inference charts. A good chart can reconstruct the same invariant while still assigning a different conventional coordinate when its response model differs.

Tensions become structured data

A tension is no longer only a nuisance to be averaged away. The pattern of disagreements across probes contains information about the direction δs . Probes with different W rows act like differently oriented detectors of the same hidden deformation.

Probe design becomes algebraic

The best collection of probes is not merely the collection with the smallest individual error bars. It is the collection whose W rows span the channel space. Complementary pipelines can reconstruct deformations that no single probe can identify.

11. Reconstructing the hidden channel deformation

Stack the shifts from many pipelines:

$$\delta \hat{\theta} = W \delta s$$

If W has sufficient rank, the channel deformation can be estimated:

$$\delta \hat{s} = W^* \delta \hat{\theta}$$

The pseudoinverse automatically reconstructs only the combinations that the probes can observe. Directions in the null space remain dark to that experiment set.

This gives a direct empirical programme: derive K from the Unified Mechanics action, compute W from official likelihoods, fit one δs using a subset of probes, and predict the rest.

12. What would count as a successful result?

A successful analysis would show all of the following:

- The channel response templates K are derived before looking at the target discrepancies.
- The same δs explains multiple independent pipelines.
- The inferred deformation agrees with the dynamics of the local realization action.
- Withheld probes are predicted with correct sign, magnitude, redshift dependence, and covariance.

- The result remains stable under reasonable analysis choices and alternative implementations.

13. What would falsify it?

- Each dataset requires its own unrelated deformation.
- Official response templates produce shifts inconsistent with the observed ordering.
- The inferred channel state violates conservation or the field equations.
- Adding an independent probe makes the reconstructed deformation unstable rather than better determined.
- The apparent relation disappears when full covariances, priors, nuisance parameters, and calibration uncertainties are included.

14. Relationship to Rationality, Realization, and Observation

Compressed object	Traditional meaning
R — Rationality	Determines which distinctions the pipeline can represent and which parameter combinations remain degenerate or invisible.
ϕ — Realization	Produces the physical channel response Kδs in propagation, geometry, growth, or calibration.
O _p — Observation	Uses data, covariance, priors, and model derivatives to convert that response into a conventional reported coordinate.

$$O_p \, \phi R \mapsto F_p^{-1} J_p^T C_p^{-1} K_p$$
$$\Sigma_p \, q = (O_p - O_q) \phi R$$

The compact expression and the conventional statistical expression are the same operation at different resolutions.

15. Conclusion

The proposal does not say that every cosmological tension is automatically explained by observation. It says that observation must be modeled as part of the physics whenever different pipelines convert different mixtures of propagation, geometry, matter, and calibration into the same named parameter.

The mathematical result is exact at first order: a universal physical channel deformation becomes a pipeline-dependent parameter shift through the likelihood projection $F^{-1} J^T C^{-1} K$. The scientific question is then sharply posed. Can Unified Mechanics derive K and δs in a way that predicts the observed cross-probe pattern without dataset-specific fitting?

FINAL STATEMENT

One universe can possess one invariant history while different observation systems report different coordinates of that history. The disagreement is not arbitrary; it is the measurable geometry of the maps between physical realization and inference.

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